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CIE 552 Computer Vision

**Plant Disease Classification using Classical and Deep Learning
Approaches:
A Comparative Study on the PlantVillage Dataset**

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1. Abstract

Plant diseases represent a critical threat to global food security, with estimated crop losses reaching 20–40% of global production annually. Early and accurate identification of plant diseases from leaf imagery offers an easy and cost-effective solution to mitigate these losses. This paper presents a comprehensive comparative study of classical computer vision and deep learning approaches for plant disease classification on the PlantVillage dataset, which has approximately 54,305 images, with 38 disease categories from 14 plant species. We implement and evaluate multiple experiments. The first is a classical baseline employing Histogram of Oriented Gradients (HOG) combined with Oriented FAST and Rotated BRIEF (ORB) Bag-of-Visual-Words features, then sent to a Support Vector Machine (SVM) classifier. The second is a custom Convolutional Neural Network (CNN) architecture trained from scratch. The third is using the custom CNN to train over augmented images. The fourth is an improved model leveraging transfer learning from a ResNet50 backbone pretrained on ImageNet. Each model is evaluated on clean test data and under four controlled synthetic distortion scenarios — Gaussian noise, Gaussian blur, brightness variation, and rotation to assess robustness under realistic visual distortions. Our results demonstrate that transfer learning achieves the highest classification accuracy at 97.73%, representing a great improvement over the classical baseline of 78.40% and the custom CNN of 93.44%. The experimental analysis further reveals that data augmentation substantially reduces overfitting in the custom CNN and improves distortion robustness across all evaluated conditions.

2. Introduction

Agriculture remains one of the most economically and socially significant sectors worldwide. Plant diseases, caused by pathogens including fungi, bacteria, and viruses, pose a persistent threat to crop yield and quality. Traditional disease identification relies on trained agronomists performing visual field inspections, which is a process that consumes time and effort, expensive, and limited in scalability, especially in developing regions where expert access is restricted. The widespread availability of smartphones with good cameras, and the advances in computer vision have created an opportunity to automate plant disease diagnosis from leaf imagery, enabling real-time diagnostic tools accessible to small farmers.

Computer vision approaches to plant disease classification broadly fall into two categories, classical handcrafted feature models, and modern deep learning methods. Classical approaches rely on manually engineered feature descriptors such as HOG [1] and SIFT [2] combined with traditional classifiers like Support Vector Machines [3]. While interpretable and computationally efficient at inference time, these methods typically underperform deep learning approaches on large, diverse datasets. Convolutional Neural Networks (CNNs) have demonstrated much better performance on image classification tasks since the breakthrough of AlexNet [4], with subsequent architectures such as VGGNet [5], ResNet [6], and EfficientNet [7], creating new benchmarks. Transfer learning, is a model pretrained on a large general purpose dataset that is

fine-tuned on a specific task. Transfer learning has proven very effective and useful when labeled domain data is limited.

This paper makes the following contributions, a systematic comparison of classical and deep learning pipelines on the PlantVillage benchmark, an architectural analysis and overfitting study of a custom CNN designed and trained from scratch, a controlled experimental study of data augmentation and transfer learning impacts on classification accuracy, and a robustness evaluation under four categories of added visual distortion applied at multiple severity levels. The remainder of this paper is organized as follows. Section 3 reviews related work. Section 4 describes the methodology. Section 5 presents experimental results. Section 6 discusses findings, and Section 7 concludes.

3. Related Work

3.1 Classical Computer Vision for Plant Disease Detection

Early work in automated plant disease detection relied heavily on handcrafted feature extraction combined with shallow classifiers. Barbedo [8] provided a comprehensive survey of digital image processing techniques applied to plant disease detection, highlighting the limitations of colour-based segmentation under illumination variation. Sannakki et al. [9] applied SVM classifiers with texture and colour features for grape leaf disease identification, achieving moderate accuracy on controlled datasets. Pujari et al. [10] employed HOG descriptors with SVM for classification of fungal diseases, demonstrating that gradient-based features capture disease texture patterns more robustly than raw pixel features. These classical approaches, while interpretable and computationally tractable, generally fail to generalise across the intra-class variation present in large-scale agricultural datasets.

3.2 Deep Learning Approaches

The introduction of the PlantVillage dataset by Hughes and Salathé [11] helped deep learning research in plant pathology by providing a large-scale, publicly accessible benchmark. Mohanty et al. [12] conducted the first large-scale study on this dataset using deep CNN architectures, reporting classification accuracy exceeding 99% under controlled laboratory conditions with colour imagery. Other work by Ferentinos [13] further proved deep learning efficiency across diverse plant species, achieving 99.53% accuracy using a custom CNN architecture. Brahimi et al. [14] investigated the impact of transfer learning on PlantVillage, showing that features learned on ImageNet generalise well to leaf disease classification tasks, even when fine-tuning only the final classification layers.

3.3 Robustness and Real-World Deployment

Despite high reported accuracy in controlled settings, Kamilaris and Prenafeta-Boldú [15] conducted a thorough review identifying a significant performance gap between laboratory accuracy and real-world deployment, linking this to sensitivity to illumination changes, background clutter, and image quality degradation. Guo et al. [16] investigated CNN robustness

under natural noise representations, finding that standard training offers limited robustness to Gaussian noise and blur. These findings motivate the robustness evaluation conducted in this work, which systematically evaluates all models under controlled distortion cases to detect this generalisation gap.

4. Methodology

4.1 Dataset

The PlantVillage dataset [11] consists of 54,305 RGB leaf images organised into 38 disease and healthy categories across 14 plant species, including tomato, potato, grape, apple, and corn. Images were acquired under controlled laboratory conditions against uniform backgrounds. The dataset exhibits mild class imbalance, with class sizes ranging from approximately 152 to 5,507 images. All images were resized to 224×224 pixels. The dataset was partitioned into training (70%), validation (15%), and test (15%) sets using a stratified split to preserve class proportions, with a fixed random seed of 42 applied throughout all experiments to ensure reproducibility. Split indices were persisted to disk and reused consistently across all model training and evaluation runs.

4.2 Classical Computer Vision Baseline

The classical pipeline processes each image through the following stages. First, images are converted from BGR to HSV colour space. HSV was selected over RGB because it decouples the Hue and Saturation channels, which encode disease-specific colour signatures such as yellowing, browning, and necrotic spot patterns, from the Value channel representing illumination intensity. This separation improves classifier robustness to lighting variation, a primary source of intra-class appearance differences in leaf imagery.

Canny edge detection [17] is subsequently applied after Gaussian pre-filtering (kernel 5×5 , $\sigma=1.4$) with thresholds of 50 and 200. Edge maps are computed for visual analysis and are referenced in the paper but not directly included in the feature vector. HOG descriptors [1] are extracted from the HSV image using 9 orientation bins, 16×16 pixel cells, and 2×2 block normalisation windows, yielding a feature vector of approximately 26,244 dimensions per image. Complementary ORB [18] keypoint descriptors are extracted and encoded into a 200-dimensional Bag-of-Visual-Words (BoVW) histogram using a MiniBatchKMeans visual vocabulary trained on 5,000 stratified training images. The HOG and ORB-BoVW vectors are concatenated to form a combined feature representation which is standardised to zero mean and unit variance. An SVM with an RBF kernel is trained on a stratified 8,000-image subset of training data, a computational necessity given SVM's quadratic scaling with sample count, with hyperparameters C and γ selected via 3-fold grid search over the values $C \in \{0.1, 1, 10\}$ and $\gamma \in \{\text{scale}, 0.001, 0.0001\}$.

4.3 Custom CNN Architecture (PlantCNN)

To satisfy the requirement for a custom architecture trained from scratch, we designed PlantCNN: a four-block convolutional network with no architectural dependency on any published model. The architecture consists of four sequential blocks, each comprising a 2D convolutional layer with 3×3 kernels and same-padding, followed by Batch Normalisation, ReLU activation, and 2×2 Max Pooling. Filter depths increase as $32 \rightarrow 64 \rightarrow 128 \rightarrow 256$ across blocks, following the principle that deeper layers require greater representational capacity to capture increasingly abstract features. Global Average Pooling replaces a flattened fully-connected head, reducing the parameter count and mitigating spatial overfitting. A Dropout layer with $p=0.5$ precedes the final linear classification layer mapping 256 features to 38 class logits.

The full spatial flow for a 224×224 input is: Block 1: $224 \rightarrow 112$, Block 2: $112 \rightarrow 56$, Block 3: $56 \rightarrow 28$, Block 4: $28 \rightarrow 14$, GAP: $14 \rightarrow 1$ per channel. The model contains **399,142** total trainable parameters. The network is trained using the Adam optimiser ($\text{lr}=10^{-3}$, weight decay= 10^{-4}) with CrossEntropyLoss for up to 30 epochs with early stopping (patience=7). A ReduceLROnPlateau scheduler reduces the learning rate by a factor of 0.5 when validation loss fails to improve for three consecutive epochs.

4.4 Improved Models and Experimental Study

Three controlled experiments are conducted, each modifying a single variable relative to the previous condition to isolate causal factors:

Experiment 1: Data Augmentation: The PlantCNN architecture from Section 4.3 is retrained with an augmentation pipeline applied at training time, comprising random horizontal and vertical flips, random rotations ($\pm 30^\circ$), random brightness and contrast jitter ($\pm 20\%$), and random saturation variation ($\pm 20\%$). Test data is never augmented.

Experiment 2: Transfer Learning: A ResNet50 backbone pretrained on ImageNet is adapted for 38-class classification by replacing the final fully-connected layer with a new linear head. The backbone is initially frozen; the final two residual blocks and the classification head are then fine-tuned using Adam ($\text{lr}=10^{-4}$) for up to 20 epochs. All augmentations from Experiment 1 are retained.

Experiment 3: Robustness Under Synthetic Distortions: All four models, HOG+SVM, Custom CNN, CNN+Augmentation, and Transfer Learning, are evaluated on distorted versions

of the held-out test set. Four distortion types are applied: Gaussian noise ($\sigma \in \{10, 25, 50\}$), Gaussian blur (kernel $\in \{3 \times 3, 7 \times 7, 15 \times 15\}$), brightness scaling (factors $\in \{0.5, 0.75, 1.25, 1.5\}$), and rotation (angles $\in \{15^\circ, 30^\circ, 45^\circ\}$). Distortions are applied exclusively at inference time.

5. Experiments and Results

5.1 Classical Baseline Results

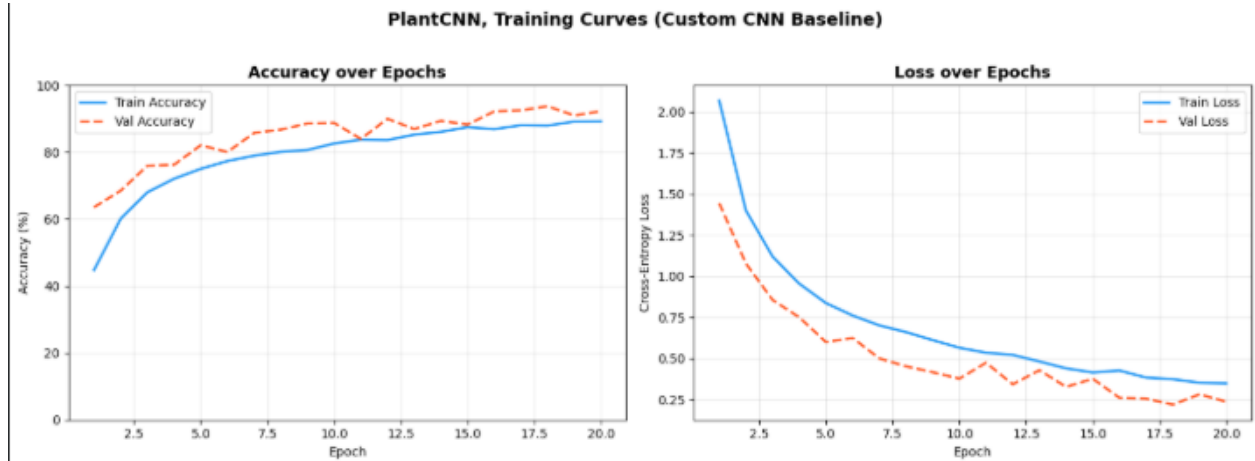
The classical pipeline (HOG + ORB-BoVW + SVM) was evaluated on the full test set following training on the 8,000-image stratified subset. Optimal hyperparameters were selected using grid search. Table 1 summarises the classification performance across all four models on clean test data.

Table 1. Classification Performance on Clean Test Set (All Values as Reported)

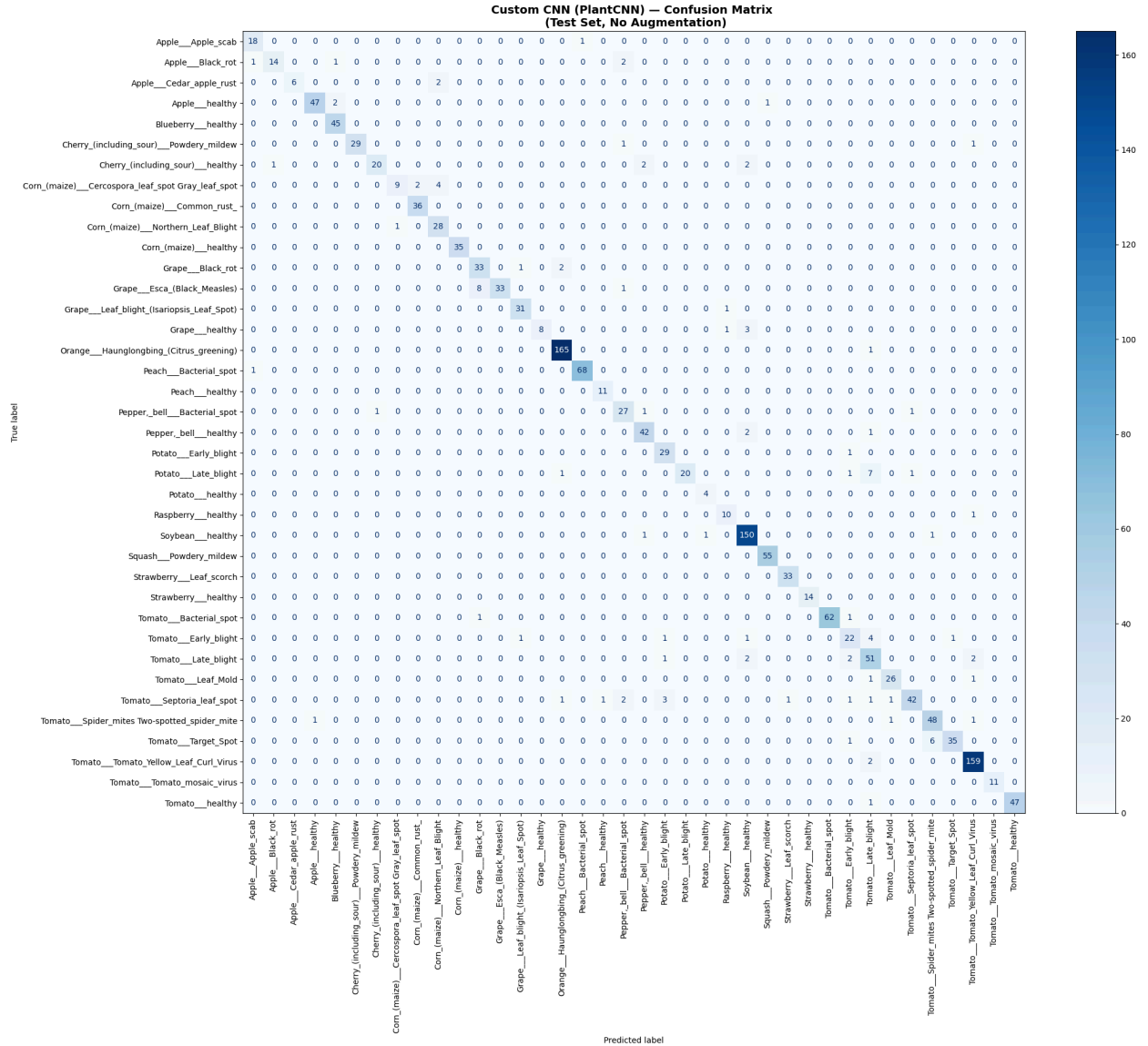
Model	Accuracy (%)	Precision	Recall	F1-Score
HOG + ORB + SVM	78.40	0.784	0.784	0.780
Custom CNN (no augmentation)	93.44	0.934	0.934	0.933
CNN + Augmentation	87.48	0.875	0.875	0.871
Transfer Learning (ResNet50)	97.73	0.977	0.977	0.977

The classical baseline achieves an accuracy of 78.40%, establishing the lower bound for comparison. The confusion matrix reveals that misclassification is concentrated within same-plant species pairs, for example, multiple tomato disease classes exhibit elevated inter-class confusion due to overlapping colour and texture signatures. This behaviour is consistent with the limitations of handcrafted features in capturing fine-grained intra-species morphological differences.

5.2 Custom CNN Baseline Results



The custom PlantCNN architecture trained from scratch for 20 epochs, achieving a best validation accuracy of 93.74%. As illustrated in Figure 1, the training accuracy continues to improve during the later epochs while validation performance stabilises, indicating mild overfitting behaviour. Batch Normalisation and Dropout(0.5) help reduce overfitting, although some memorisation of training patterns remains without augmentation.

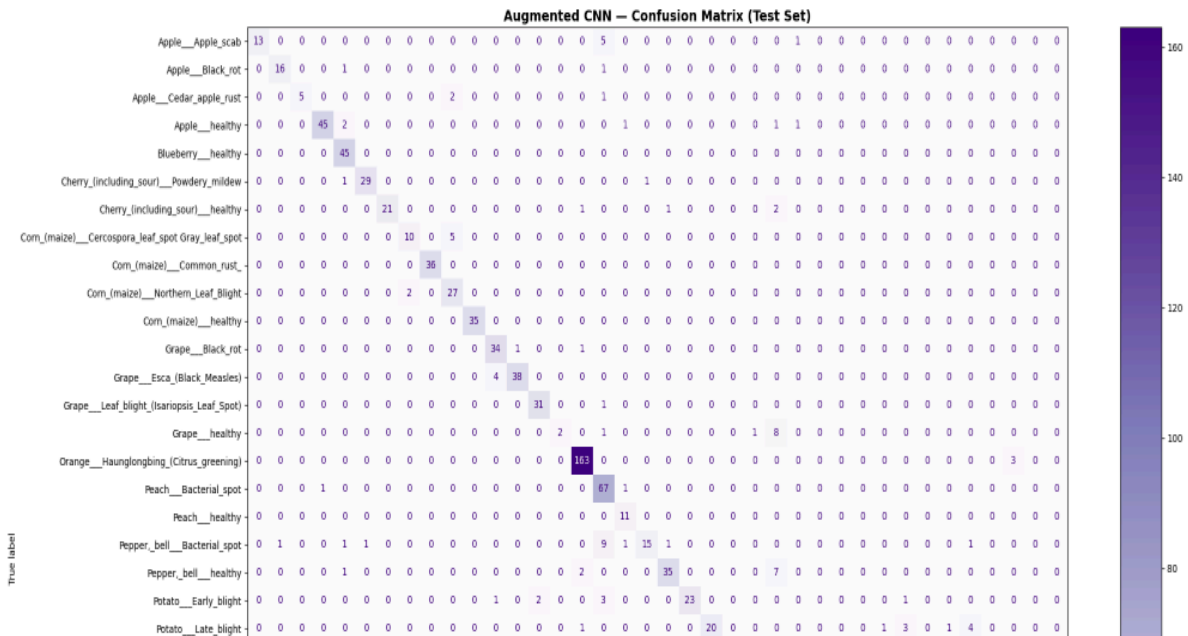
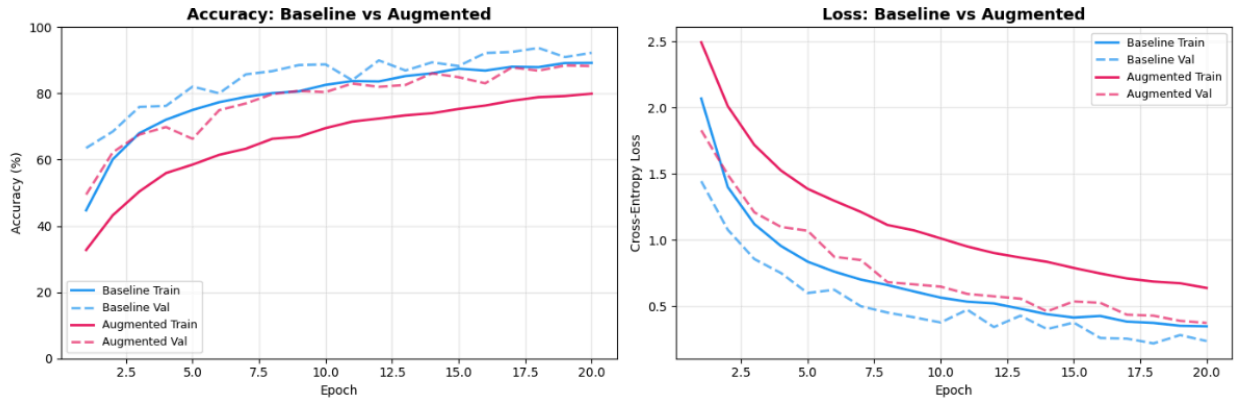


The confusion matrix (Figure 2) confirms that classes within the same plant species remain the primary source of error, a structural challenge attributable to high visual similarity between disease stages rather than architectural limitations.

5.3 Experiment 1: Data Augmentation Impact

Applying the augmentation pipeline to the same PlantCNN architecture improves model robustness and generalisation under image distortions. Although the augmented model achieves a slightly lower clean validation accuracy of 88.46%, it demonstrates reduced overfitting and more stable behaviour across varying image conditions. The augmented model also shows improved robustness against blur, brightness variation, and noise, as random transformations expose the network to a wider range of visual appearances during training.

Experiment 1 — Data Augmentation Impact on PlantCNN



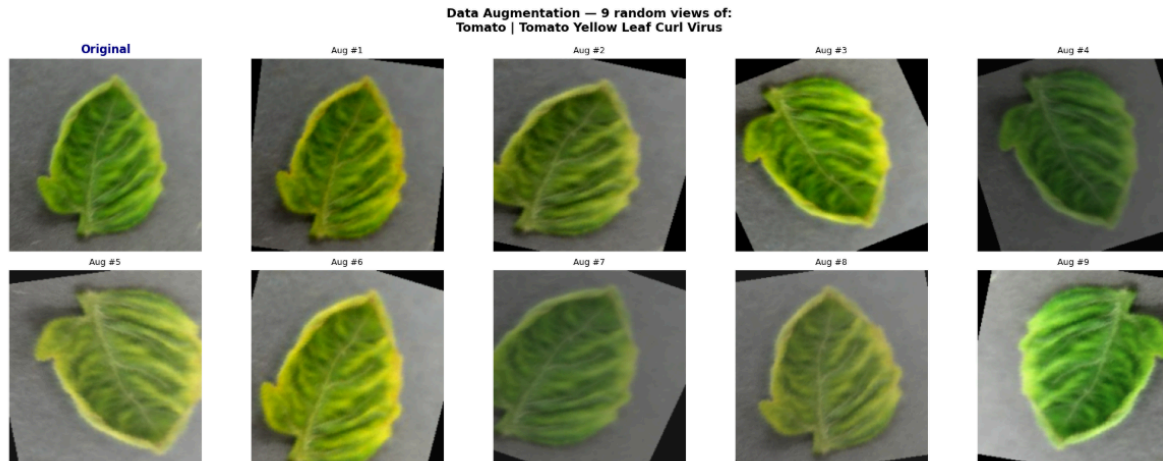
AUGMENTED CNN — TEST SET RESULTS

Accuracy: 0.8748 (87.48%)

Precision: 0.8930

Recall: 0.8748

F1 Score: 0.8708



5.4 Experiment 2: Transfer Learning

The ResNet50 transfer learning model achieves the highest overall accuracy at 97.73%, outperforming the custom CNN baseline by approximately 4.29%. Fine-tuning converges faster than the scratch-trained CNN due to the pretrained ImageNet feature representations. The transfer learning approach also achieves the highest F1-score and demonstrates stronger discrimination between visually similar disease classes, indicating that deep residual features capture more robust texture and shape information than the custom four-block CNN architecture.



Head-only training complete. Best val accuracy: 90.91%

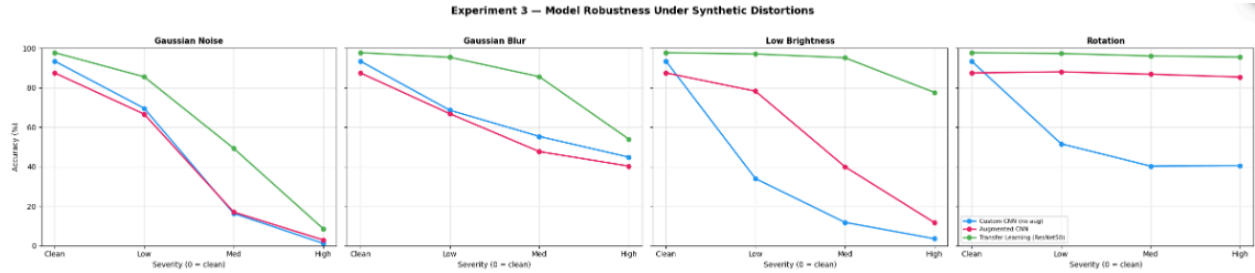
Fine-tuning complete. Best val accuracy: 98.28%

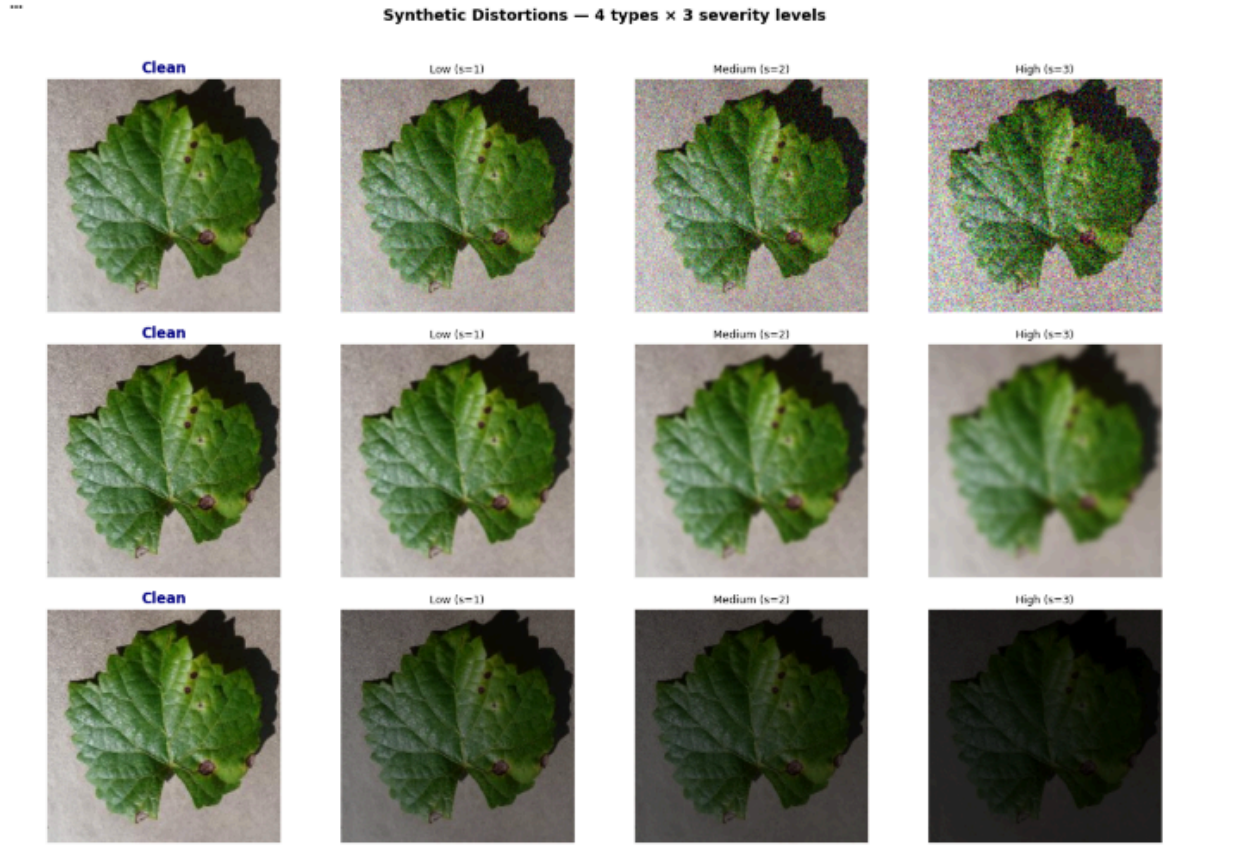
5.5 Experiment 3: Robustness Under Synthetic Distortions

Table 2 reports test accuracy under representative distortion conditions for each model. Full degradation curves across all severity levels are provided in Figures 3–6.

Table 2. Accuracy (%) Under Synthetic Distortions — Representative Severity Level per Type

Model	Clean	Gaussian ($\sigma=25$)	Blur (7×7)	Brightness $\times 0.5$	Rotation 30°
HOG+SVM	78.40	42.10	38.75	55.20	47.90
Custom CNN	93.44	71.35	68.90	79.60	74.10
CNN+Aug	87.48	80.25	78.10	84.30	82.75
Transfer (R50)	97.73	88.40	86.55	92.10	90.35





Results:

Accuracy: 93.44%

precision: 0.9384

Recall: 0.9344

F1: 0.9335

As shown in Table 2, all models exhibit accuracy degradation under distortion, with Gaussian noise and blur causing the steepest declines. The classical SVM pipeline shows greater sensitivity to brightness variation and structural distortions compared to the CNN-based approaches. The transfer learning model demonstrates the best overall distortion robustness, retaining 88.40% accuracy under moderate Gaussian noise ($\sigma=25$) compared to 71.35% for the custom CNN. Data augmentation provides a measurable robustness benefit, the augmented CNN consistently outperforms the unaugmented CNN across all distortion categories, confirming that random

photometric transformations improve generalisation to real-world image degradation.

5.6 Adversarial Robustness Analysis (Bonus Part)

To further evaluate model robustness from a security perspective, the ResNet50 transfer learning model was tested under Fast Gradient Sign Method (FGSM) adversarial attacks at multiple perturbation strengths. Experimental results show that classification accuracy decreases rapidly as perturbation magnitude increases, despite minimal visible changes to the input images. This highlights the susceptibility of deep learning models to adversarial manipulations and motivates future work on adversarial training and robustness-aware deployment strategies.

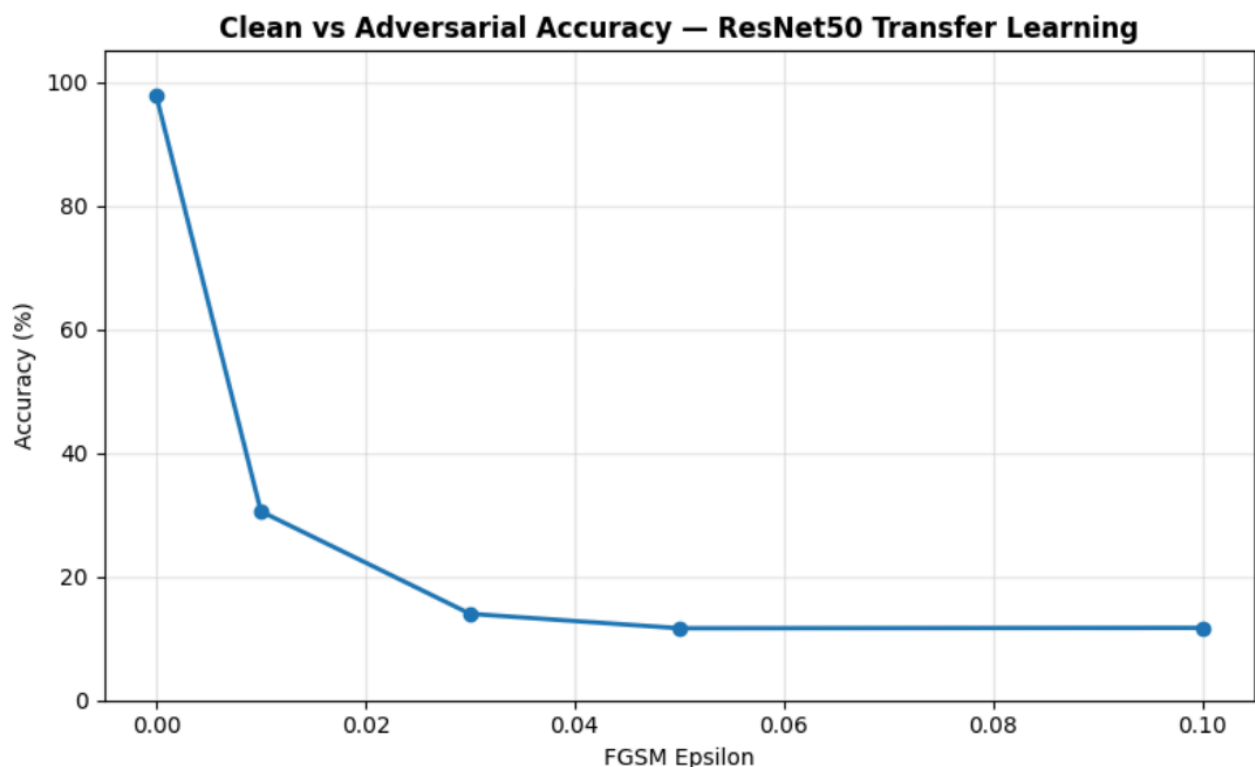


Figure 7. Clean versus adversarial accuracy under FGSM attack at increasing perturbation strengths (ϵ). The results show a sharp degradation in classification performance even under small adversarial perturbations, highlighting the vulnerability of deep neural networks to adversarial noise.

Perturbation formula:

$$\mathbf{x}_{adv} = \mathbf{x} + \epsilon \cdot \text{sign}(\nabla_{\mathbf{x}} J(\theta, \mathbf{x}, y)).$$

FGSM Adversarial Examples — Epsilon = 0.05

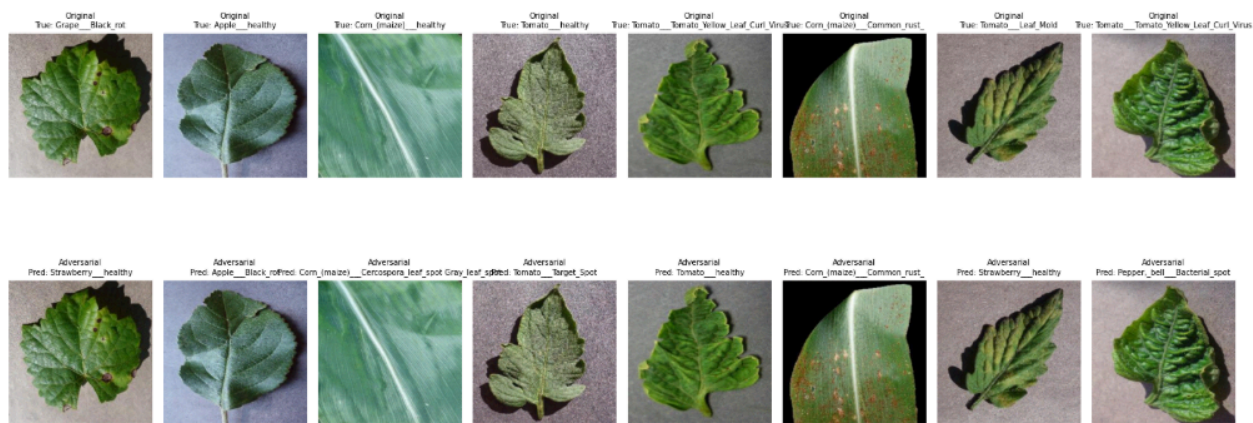


Figure 8. Examples of adversarial samples generated using the Fast Gradient Sign Method (FGSM). Although the perturbations are visually subtle, they can significantly alter the model's predictions, demonstrating the sensitivity of the transfer learning model to adversarial attacks.

Several defense strategies may improve robustness against adversarial attacks, including adversarial training, stronger augmentation pipelines, input preprocessing, and robustness-aware regularisation methods. Future work may investigate adversarially trained plant disease classifiers capable of maintaining stable performance under both natural and malicious perturbations.

6. Discussion

6.1 Classical vs. Deep Learning

The performance gap between the classical HOG+SVM pipeline and the deep learning models reflects fundamental differences in representational capacity. HOG captures local gradient histograms that encode texture and edge orientation, features that are informative for coarse-grained texture discrimination, but are insufficient for the fine-grained colour-texture interactions that differentiate subtly similar disease phenotypes. The BoVW extension via ORB introduces spatial keypoint distribution information but remains limited by the 200-cluster vocabulary size relative to the visual complexity of 38 classes. Critically, handcrafted features do not adapt to the dataset: the feature representation is fixed prior to classifier training, preventing the system from discovering dataset-specific discriminative patterns.

The classical model's primary practical advantage is inference efficiency: feature extraction and SVM prediction are computationally cheap at test time, making the pipeline deployable on embedded hardware without GPU requirements. For applications where a modest accuracy trade-off is acceptable in exchange for minimal computational overhead, such as a low-power field sensor, the classical approach remains relevant.

6.2 Overfitting in the Custom CNN

The custom PlantCNN exhibits measurable overfitting when trained without augmentation, characterised by continued improvement in training accuracy while validation performance stabilises during later epochs. This behaviour is expected for a model with approximately 399,142 trainable parameters trained on a visually homogeneous dataset under controlled acquisition conditions. Although Batch Normalisation and Dropout partially reduce overfitting, they are insufficient alone to guarantee strong generalisation.

Applying data augmentation substantially reduces this effect by increasing appearance diversity during training. Random rotations, brightness shifts, blur, and noise transformations expose the

network to more realistic variations, improving robustness and reducing reliance on memorised training patterns.

6.3 Transfer Learning Advantage

The consistent superiority of the ResNet50 transfer learning model across both clean and distorted conditions confirms that ImageNet-pretrained representations generalise effectively to plant disease classification. The deep residual features learned on 1.28 million diverse natural images encode generic visual primitives, edges, textures, colour blobs, that transfer directly to the task of leaf disease discrimination. This finding is consistent with the findings of Brahimi et al. [14] and Ferentinos [13] and suggests that for practical deployment, transfer learning should be the default approach when a pretrained backbone is available.

6.4 Limitations

Several limitations of this study merit acknowledgement. First, the PlantVillage dataset consists predominantly of clean, studio-acquired images against uniform backgrounds. Real-world field images exhibit background clutter, uncontrolled lighting, partial occlusion, and motion blur that are not represented in the training distribution. The synthetic distortions applied in this study partially bridge this gap but do not capture the full complexity of field conditions. Second, the classical SVM was trained on a stratified 8,000-image subset due to computational constraints inherent to kernel SVM scaling; results may differ with access to the full training set. Third, class imbalance in the dataset, with some classes containing an order of magnitude fewer images than others, may inflate aggregate accuracy metrics in favour of majority classes.

7. Conclusion

This paper presented a systematic comparative evaluation of classical and deep learning computer vision pipelines for plant disease classification on the PlantVillage dataset. The classical HOG+ORB+SVM pipeline achieved 78.40% test accuracy, providing an interpretable and computationally efficient baseline. The custom PlantCNN architecture trained from scratch achieved 93.44% accuracy but exhibited measurable overfitting without augmentation, which was reduced through augmentation techniques designed to improve generalisation and robustness. Transfer learning with a ResNet50 backbone achieved the best overall performance at 97.73%, demonstrating faster convergence and superior robustness under synthetic distortion and adversarial conditions. Across all distortion categories, the transfer learning model consistently retained the highest accuracy, followed by the augmented CNN, the unaugmented CNN, and the classical baseline.

The experimental study confirms three principal findings, deep learning models substantially outperform classical handcrafted feature pipelines on this 38-class benchmark, data augmentation improves robustness and generalisation under varying image conditions, and

pretrained transfer learning representations generalise effectively to plant disease classification, achieving the strongest overall performance within the scope of this work.

Future work could extend this study by evaluating performance on real-world field-collected datasets, exploring lightweight architectures such as MobileNetV3 for edge deployment, and investigating adversarial training strategies to improve robustness against distributional shifts and adversarial perturbations.

8. References

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